

IDENTIFICATION

CODE : GI-5-S1-EC-DSC
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

Final exam 1h50.

TEACHING AIDS

Slides (lectures) + Python Jupyter notebooks (practicals)

TEACHING LANGUAGE

English

CONTACT

M. MONCLA Ludovic :
ludovic.moncla@insa-lyon.fr

AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A5 Process data (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C13 Considering technological and methodological innovation (Level 2)

CONTENT

The Data Science course is structured into two core parts:

1. Introduction to Artificial Intelligence – This section focuses on Deep Learning and Reinforcement Learning, covering key concepts such as deep neural networks (RNN, LSTM, Transformers), activation functions, optimization techniques, and loss functions. It also explores real-world applications, including self-driving car simulations and AI-driven decision-making.

2. Introduction to Natural Language Processing (NLP) – This section covers the foundations and applications of Large Language Models (LLMs). Topics range from the history of NLP techniques to modern industrial applications in Customer Support, Legal Tech, and Industry 4.0. The course covers essential techniques such as text classification, word/document representation, and Retrieval-Augmented Generation (RAG).

The course includes hands-on labs using Python and Jupyter notebooks, providing practical experience in AI and NLP model development.

BIBLIOGRAPHY

Charu C. Aggarwal. (2023). Neural networks and deep learning. A Textbook. Switzerland: Springer International Publishing.

Nithin Buduma, Nikhil Buduma, Joe Papa (2022) Fundamentals of Deep Learning, 2nd Edition.

Laura Graesser and Wah Loon Keng. (2019) Foundations of Deep Reinforcement Learning Theory and Practice in Python. Addison-Wesley Professional.

Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition.

PRE-REQUISITES

Python programming

Probability theory

Introduction to Data Science